PRECALCULUS · UNIT 2: FUNCTIONS

Streaming Service Analyst

Use function families, transformations, compositions, and inverses to analyze a streaming business.

CP2

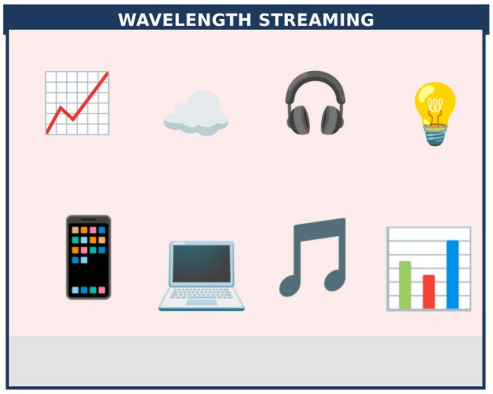
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You are a data analyst at Wavelength, a new music streaming service. Leadership wants to know how pricing plans work, how fast subscribers are growing, and when the company will reach its goals.

You will use functions to model the business and present your findings.



Subscribers

$S(t) = 20{,}000(1.08)^t$

t in months

Plans

Basic $5 · Premium $11

Family $17

Learning Goals

- I can identify domain and range and evaluate functions.
- I can describe transformations of a graph.
- I can use exponential models to make predictions.

Project Timeline

When	What you will do
Day 1	Plans as Functions
Day 2	Transformations
Day 3	Growth
Final	Analyst Briefing + reflection

Part 1: Plans as Functions

Day 1 · Basic costs $5 per month plus $2 for each GB over 10 GB.

Business Data Cards

Data Overage	Exchange	Tax
$2 per GB over 10 GB	E(d) = 0.92d	T(x) = 1.0625x
on the Basic plan	dollars → euros	

Growth	Churn	Goal
8% per month	Old app users fall	100,000 subscribers
from 20,000	5% per month from 50,000	

1. Write the cost C(g) for g GB if g ≤ 10 and if g > 10.

2. Find C(6), C(10), and C(14).

3. What is a sensible domain and range for C?

4. Is C a function? Explain.

Part 2: Transformations

Day 2 · The parent graph is $y = x^2$.

5. Describe how $y = (x - 3)^2 + 2$ is related to $y = x^2$.

6. Describe $y = -2x^2$.

7. Write the equation of $y = |x|$ shifted left 4 and down 1.

8. Basic's price rises $1 per month. How does the graph of C change?

Part 3: Growth

Day 3 · $S(t) = 20{,}000(1.08)^t$.

9. Find S(6) and S(12). Round to the nearest subscriber.

10. Use a table to find the first month the subscribers reach 100,000.

11. Is this growth linear or exponential? How do you know?

Final Product: Analyst Briefing

Your final product must include:

- Piecewise plan cost with a graph
- Transformation descriptions
- Subscriber predictions
- Month the goal is reached
- One-page briefing

Reflection

R1. How does a piecewise function model a real price plan?

R2. Why does exponential growth start slow and then speed up?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Functions, transformations, compositions, inverses, and exponential models are built and interpreted correctly. Goes beyond the minimum with a creative, realistic choice.	Functions, transformations, compositions, inverses, and exponential models are built and interpreted correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Briefing includes models, graphs, calculations, and a recommendation. Polished, organized, and visually engaging.	Briefing includes models, graphs, calculations, and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.